Assignment 5
Naïve Bayes Text Classifier

Goal:

- Build a Naïve Bayes model for text classification
- Use your Wikipedia dataset from Assignment 2
- Evaluate and interpret the model
- Understand what the model learns

Before You Begin

You must have from Assignment 2:

- final_dataset.csv
- Column: clean_text
- Column: label

👉 Do NOT redo scraping — use your existing dataset

Part 1 — Theory (Write in Markdown)

Answer these in your notebook:

1. What is **Bayes' Theorem**? (words + formula)
2. What does "naïve" mean in Naïve Bayes?
3. Why does Naïve Bayes still work well?
4. What is **Laplace smoothing**? Why is it needed?
5. Difference between:
    - MultinomialNB
    - GaussianNB
    - BernoulliNB
      👉 Which one is best for text and why?

Part 2 — Load & Explore

- Load dataset
- Show shape & first rows
- Plot class distribution
- Check if data is balanced

Part 3 — Model Building

- Split data (80–20, stratify)
- Create pipeline:
  - TF-IDF
  - MultinomialNB
- Train model
- Make predictions

Part 4 — Evaluation

- Classification report
- Confusion matrix
- Accuracy

👉 Write short observations for each

Part 5 — Model Understanding

- Top 20 words per class
- Show:
  - 5 most confident predictions
  - 5 least confident predictions
- Try smoothing (`alpha=1.0` vs `0.1`)

Part 6 — Model Comparison

Train and compare:

- MultinomialNB
- BernoulliNB
- ComplementNB

👉 Compare accuracy & F1 score

Final Submission

1. GitHub Repository must include:

- `naive_bayes_classifier.ipynb`
  👉 Clean, well-structured, and clearly explained
- `README.md`
  👉 Simple explanation including:
    - Dataset used (from Assignment 2)
    - Approach (TF-IDF + Naïve Bayes)
    - Models tried (Multinomial, Bernoulli, Complement)
    - Final results (best model + accuracy/F1 score)
    - Key insights

👉 Do NOT upload the dataset again — reuse or provide link to previous repository

2. Write-up (150–250 words)

Post on LinkedIn or include in README:

- What problem you solved (text classification)
- Which Naïve Bayes model you used
- Your best result (accuracy / F1 score)
- One interesting insight (e.g., important words per class)
- One surprising thing you learned

3. Submit Linkedin and Github repo link on LMS

💡 Tips for a Strong Submission

- Use **markdown cells** to explain:
    - Your model choice
    - Results and observations
- Visualise:
    - Class distribution
    - Confusion matrix
    - Top words per category
- Keep your notebook:
    - Clean
    - Well-structured
    - Error-free (restart kernel and run all cells before submitting)
- Your README should:
    - Be simple and easy to understand
    - Explain your project clearly to a beginner

- The write-up is what gets attention 🚀
  👉 Spend time making it clear and insightful